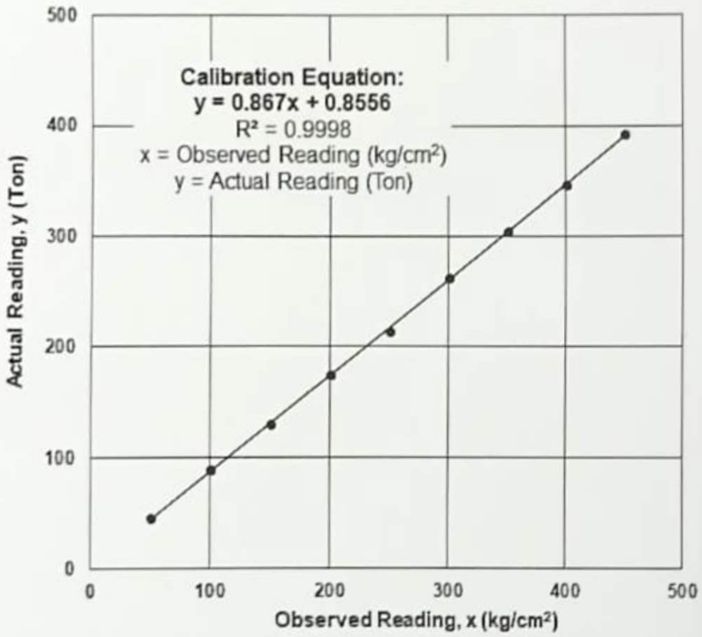
CONSULTANCY RESEARCH & TESTING SERVICES (CRTS)
Department of Civil Engineering
Khulna University of Engineering & Technology
Khulna-9203, Bangladesh
Phone: +8802477733351-70 Ext. 201; Cell: +88-01798-238240, 01714-087452

কনসালটেন্সি রিসার্চ এন্ড টেস্টিং সার্ভিসেস (সিআরটিএস)
সিভিল ইঞ্জিনিয়ারিং বিভাগ
খুলনা প্রকৌশল ও প্রযুক্তি বিশ্ববিদ্যালয়
খুলনা-৯২০৩, বাংলাদেশ
http://www.kuet.ac.bd
Email: crts-civil@kuet.ac.bd

CALIBRATION OF HYDRAULIC JACK WITH PRESSURE GAUGE

CRTS No. : CRTS/Civil/R-26327 — Date : 06-07-2026
Ref. No. : -- — Date : 06-07-2026
Client : মেসার্স মাসুম ট্রেডার্স, রহনপুর, গোমস্তাপুর, চাঁপাইনবাবগঞ্জ
Name of the Project : --
Jack Information : Hydraulic Jack (ID: L709047); Jack Capacity: 500 Ton; External Diameter of Rammer: 30.1 cm; Body Diameter: 46.0 cm; Body Height: 44.5 cm
Pump Information : Pump ID: 710567; Pump Capacity: 500 Ton
Meter Information : Range:0-600 kg/cm^2; Meter ID: 256050C
Test No. : T-262174 — Date of Calibration : 08-07-2026



Serial No.	Observed Reading	Calibrated Reading
	x (kg/cm^2)	y (Ton)
1	50	44.2
2	100	87.6
3	150	130.9
4	200	174.3
5	250	217.6
6	300	261.0
7	350	304.3
8	400	347.7
9	450	391.0

Disclaimer: This calibration is valid only when the above-mentioned Hydraulic Jack and Pressure Gauge pair is used together as they were used during calibration test. Re-calibration is required if any of the above Jack and / or Pressure Gauge is changed/replaced or repaired.

Countersigned by

Chairman
CRTS
Department of Civil Engineering
Khulna University of Engineering & Technology

T ID=T-262174

Calibrated by

Mohei Menul Islam
Assistant Professor
Department of Civil Engineering
Khulna University of Engineering & Technology

Notes:
- CRTS (Civil) does not have any responsibility regarding the representative character of the samples supplied by the client. It is recommended that samples be sent in a secure and sealed cover/packet/container under signature of the competent authority.
- In order to avoid fraudulent fabrication of test results, it is recommended that all test reports should be collected by person duly authorized.
- For any query, please contact with Chairman, CRTS (Civil), Department of Civil Engineering, KUET.

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CRTS No. : CRTS/Civil/R-26327 — Date : 06-07-2026
Ref. No. : -- — Date : 06-07-2026
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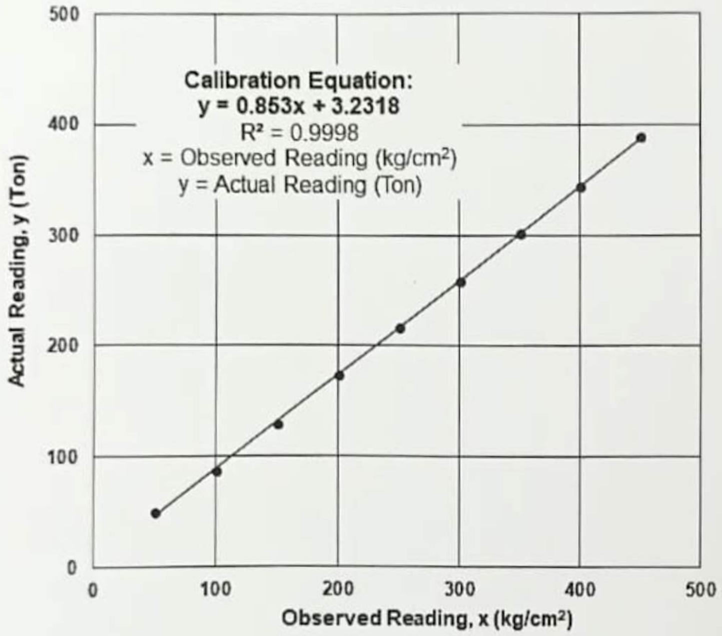
Name of the Project : --
Jack Information : Hydraulic Jack (ID: L709046); Jack Capacity: 500 Ton; External Diameter of Rammer: 30.1 cm; Body Diameter: 46.0 cm; Body Height: 44.5 cm
Pump Information : Pump ID: 710570; Pump Capacity: 500 Ton
Meter Information : Range:0-600 kg/cm²; Meter ID: 237227E
Test No. : T-262174 — Date of Calibration : 08-07-2026



Serial No.	Observed Reading	Calibrated Reading
	x (kg/cm²)	y (Ton)
1	50	45.9
2	100	88.5
3	150	131.2
4	200	173.8
5	250	216.5
6	300	259.1
7	350	301.8
8	400	344.4
9	450	387.1

Disclaimer: This calibration is valid only when the above-mentioned Hydraulic Jack and Pressure Gauge pair is used together as they were used during calibration test. Re-calibration is required if any of the above Jack and / or Pressure Gauge is changed/replaced or repaired.

DEPARTMENT OF CIVIL ENGINEERING • CRTS • KUET

Countersigned by

Chairman
CRTS
Department of Civil Engineering
Khulna University of Engineering & Technology

Calibrated by

Mohei Menul Islam
Assistant Professor
Department of Civil Engineering
Khulna University of Engineering & Technology

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09.07.20

Construction of 301m long PSC Girder Bridge
UZ: Pirgaonj, Dist: Rangpur

Design Load = 3750 kN ≅ 382.394 Ton [1 ton = 9.80665 kN]

Grip length = 580 mm

Jack ID: L709047

Pump ID: 710567

$Y = 0.867x + 0.8556$

$X = \dfrac{Y - 0.8556}{0.867}$

$X_{20} = 87.22$

$x_{40} = 175.43$

$x_{60} = 263.65$

$x_{80} = 351.86$

$x_{95} = 418.01$

$x_{98} = 431.25$

$x_{100} = 440.07$

$x_{102} = 448.89$

$x_{105} = 462.12$

Jack ID: L709046

Pump ID: 710570

$Y = 0.853x + 3.2318$

$X = \dfrac{Y - 3.2318}{0.853}$

$x_{20} = 85.87$

$x_{40} = 175.53$

$x_{60} = 265.19$

$x_{80} = 354.85$

$x_{95} = 422.09$

$x_{100} = 444.50$

$x_{102} = 453.47$

$x_{105} = 466.92$

Design Data:

Area of strand = 140 mm²

Area of cable A_1 = 2660 mm²

Modulus of Elasticity of cable E_1 = 197000 MPa

Avg slip = 6 mm

Test Data:

Area of strand = 142.50 mm²

Area of cable A_2 = 2707.50 mm²

Modulus of Elasticity of cable E_2 = 200400 MPa

Elongation for Grip length :

$$\delta = \frac{PL}{AE}$$

$$= \frac{3750 \times 580}{2660 \times 197000}$$

$$= 4.15 \text{ mm}$$

Correction factor :

$$\frac{A_1 E_1}{A_2 E_2} = \frac{2660 \times 197000}{2707.50 \times 200400}$$

$$= 0.966$$

Cable No	Design			Correction factor	Corrected Elongation	Corrected Elongation for Grip length
	Elongation	Elongation for Grip length	Total Elongation			
4	150	4.15	154.15	0.966	148.91	4.01
1	142		146.15		141.18	
2	146		150.15		145.04	
3	150		154.15		148.91	